Laboratory work 04

Objectice – susipažinti su konteinerių valdymo platforma Docker.

Software – portal.azure.com; Docker engine, git, draw.io

Task

1. **VM preparation**

1.1. Download terraform script (*.tf and *.sh) from moodle
1.2. Enter your subscribtion id into main.tf
1.3. Run it
1.4. Wait couple of minutes to finish install of additional packets

2. **Getting to know Docker and containers**

2.1. Check if Docker installed

```
sudo docker version
```

2.2.
2.3. Start the first Docker container:

```
sudo docker run hello-world
```

You should see a similar output:

```
Hello from Docker!

This message shows that your installation appears to be working correctly.

< .. >
```

2.4. Let's find the Ubuntu container in the Docker registry:

```
sudo docker search ubuntu
```

2.5. Operational containers can be checked in the following way:

```
sudo docker container ls
```

The list should now be empty.

2.6. You can download (pull) and run the container:

```
sudo docker image pull ubuntu
```

```
sudo docker container run ubuntu
```

What do you see?

2.7. Or simply start the container (the system will download the desired container if it is not available locally):

```
sudo docker container run busybox
```

This time the process was different. What was different?

Have you noticed that in both cases the containers terminate automatically (stop working) after the container is started?

2.8. Check that the containers are operational

```
sudo docker container ls
```

Add an additional argument to the command

```
sudo docker container ls -a
```

What's the differenc?

2.9. This time, try adding an additional parameter:

```
sudo docker container run -it busybox
```

What do you see?

What do the -i and -t arguments do?

2.10. Create an **additional** terminal session to the server and check the running containers:

```
sudo docker container ls
```

What do you see?

Add an additional argument to the command

```
sudo docker container ls -a
```

What do you see?

2.11. In an additional terminal session, check how the allocated resources look:

```
sudo docker container stats
```

What do you notice?

3. Container ephemerality

3.1. Start the container:

```
sudo docker container run -it busybox
```

Create the file and make sure it is created inside the container:

```
touch test_failas

ls -la | grep test_file
```

Exit the container

```
exit
```

3.2. Start the container again:

```
sudo docker container run -it busybox
```

Is the file you created available?

```
ls -la | grep test_file
```

what happended?

4. Reaching containerised services

4.1. Start the container with additional network arguments -p 8080:80

```
sudo docker container run nginx
```

Can you access a standard web page via an external VM IP?

Draw a logical data flow diagram

5. Image creation

5.1. Create a separate directory and a *Dockerfile* with the following content

```
FROM httpd:2.4

COPY ./public-html/ /usr/local/apache2/htdocs/
```

1. Create an additional directory "public-html" within new directory
2. Place an index.html file with your desired content in it
3. Return to the directory where the *Dockerfile* is stored. Build an image:

```
docker build -t my-docker-image .
```

5.2. Launch the container from the image you created:

```
docker run -dit --name my-container1 -p 8080:80 my-docker-image
```

Note the *-dit* argument, figure out how to stop the container, find a command that can delete the container.

Can you access your web page via an external VM IP?

6. Getting to know Podman

6.1. Check if Podman installed

```
podman info
```

6.1. Run container

```
podman run -d -p 8080:80/tcp docker.io/library/httpd
```

6.2. Check images and containers

```
podman ps -a
podman images
```

6.3. Stop container

```
podman stop container_id
```

6.4.

7. Clean up

7.1. Destroy Azure resources

In your report, please provide:

- Work progress
- Answer the questions asked in the course of the work

- Questions
    - How do you build a container image?
    - What are some popular container runtimes?
    - How do you run a Podman container in the background?
    - Can Podman work with Docker images?
    - How do you build an image using Podman?